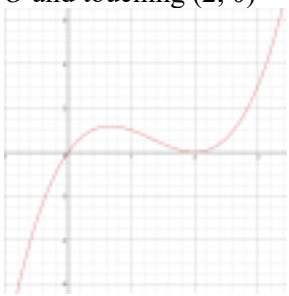
Question	Answer	Marks	Partial Marks
6(b)	74.3 or 74.30 to 74.33	**4**	**M2** for [sin ... =] $\frac{16.9\times\sin 75}{32}$ oe or **M1** for $\frac{16.9}{\sin C}=\frac{32}{\sin 75}$ oe **B1** for [angle *BCD* =] 30.7 or 30.67 to 30.69… or **M1dep** for 105 – *their* angle *BCD*
6(c)	388 or 387.7 to 387.9… nfww	**3**	**M1** for $\frac{1}{2}\times 16\times 19\times\sin 57$ oe **M1** for $\frac{1}{2}\times 16.9\times 32\times\sin$ *their* **(b)** oe
6(d)	13.4 or 13.41 to 13.42 nfww	**3**	**M2** for $\frac{x}{16}=\sin 57$ oe or **M1** for distance required is perpendicular to *AD* soi
7(a)(i)	Triangle at (4, 0) (4, 3) (6, 3)	**2**	**B1** for translation by $\begin{pmatrix}2\\k\end{pmatrix}$ or $\begin{pmatrix}k\\-1\end{pmatrix}$ If 0 scored **SC1** for triangle at (3, 0.5) (3, 3.5) (5, 3.5)
7(a)(ii)	Triangle at (1, –2) (4, –4) (4, –2)	**2**	**B1** for rotation 90 clockwise wrong centre or for rotation 90 anticlockwise about the origin
7(a)(iii)	Triangle at (–4, 4) (–4, 2.5) (–5, 2.5)	**2**	**B1** for enlargement SF $-\frac{1}{2}$ with wrong centre or for enlargement SF $\frac{1}{2}$ with centre (–2, 3)
7(b)	Reflection $y=-x$ oe	**2**	**B1** for each
8(a)	[*L* =] 11.8 [*W* =] 5.9 [*H* =] 7.1	**5**	**M1** for $L=2W$ oe soi **M1** for $W+2H=20.1$ oe **M1** for $2L+2H=37.8$ oe **B1** for at least one correct answer
8(b)(i)	0.559 to 0.56[0…]	**B4**	**M2** for $\frac{1}{3}\times 18\times 15\times\sqrt{24^2-18^2}$ isw conversion or **M1** for $h^2+18^2=24^2$ oe or better **M1** for figs 800 ÷ figs *their* volume isw
	g/cm^3 or g cm^{-3} final answer	**B1**	

Question	Answer	Marks	Partial Marks
8(b)(ii)	34.1 or 34.11 to 34.12	**4**	**M3** for tan [ ] = $\frac{\sqrt{24^2 - 18^2}}{\sqrt{18^2 + 15^2}}$ oe or **M2** for $\sqrt{18^2 + 15^2}$ isw or $\sqrt{24^2 + 15^2}$ isw or **M1** for $18^2 + 15^2$ isw or $24^2 + 15^2$ isw or **M1** for indicating required angle is *EBD*
9(a)(i)	2	**2**	**M1** for $x(x^2 - 4x + 4)$ or $x(x - 2)^2$ or $(x^2 - 2x)(x - 2)$ or $x^3 - 2ax^2 + a^2x$
9(a)(ii)	Correct sketch with curve passing through *O* and touching (2, 0) 	**4**	**B1** for any positive cubic **B1** for sketch through or touching *O* **B1** for sketch with min or max touching *x*-axis once only but not at (0, 0) **B1FT** *their* **(a)(i)** for sketch with min or max touching *x*-axis at (*their* 2, 0) and *their* 2 is labelled or clearly indicated
9(b)	$y = 20x - 64$ final answer nfww	**7**	**B6** for equivalent correct equation OR **B2** for $3x^2 - 8x + 4$ isw or **B1** for $3x^2$ or $-8x$ seen **M2dep** for [grad =] 20 soi nfww or **M1dep** for substituting 4 into *their* derivative isw **B1** for (4, 16) soi **M1dep** for 16 = *their* $20 \times 4 + c$ oe
10	125 n^3 oe final ans	**B2**	**B1** for 125 **B1** for n^3
	29 $6n - 1$ oe final ans	**B3**	**B1** for 29 **B2** for $6n - 1$ oe or **B1** for $6n + k$ or $an - 1$ ($a \neq 0$)
	2^{n-3} oe final ans	**B2**	**B1** for $2^{n\,[+\,k]}$ oe
	25 $6n - 1 - 2^{n-3}$ oe final ans OR 25.25 $-\frac{1}{24}n^3 + \frac{1}{8}n^2 + \frac{17}{3}n - 1$ oe final ans	**B2**	**FT** *their* 29 – 4 and *their* $6n - 1$ – *their* 2^{n-3} **B1FT** for each OR **B1** for each